

Mimosa MIL Experiment

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1. Motivation

In the reproduction (first report), Mimosa trained with the paper's method on the 39,770 40-nt sites of our dataset was at chance at gene level (accuracy ~ 0.52 , specificity ~ 0.14): trained only on binding sites, it fires on ordinary windows of a transcript. It also used only 1.5 % of our miRNA-mRNA pairs. This experiment keeps Mimosa's architecture and test rule and changes only the training procedure, so that all 2.4M gene-level pairs can be used.

2. Method: multiple-instance training

Mimosa calls a pair positive if at least one 40-nt window scores above 0.5. Training is made consistent with that rule: each pair is a bag of windows, the model picks the window it scores highest, and that window is trained with the pair's label. For a positive pair, at least one window must score high; for a negative pair, even its highest-scoring window must score low, so the windows that cause false positives are exactly the ones trained against. Each round has three steps:

- **Selection:** every window of every training pair is scored with the current model (no gradients); each pair keeps its best window.
- **Training:** one epoch (three in rounds 7-9) over the selected windows with the original training loop.
- **Validation:** gene-level metrics on a validation subset with the updated model.

The model is trained **from scratch**: the initial model is the one the original `perform_train` builds (same architecture, same random seed). In round 1 the best window of an untrained model is arbitrary, so a random window per pair is used instead. Nine rounds were run: rounds 1-6 with one epoch on the selection, rounds 7-9 with three epochs (optimizer state carried over), added because the training loss was still falling.

Unchanged from Mimosa	Changed
Architecture: 16 encoder blocks, 8 heads, dim 64, 40-nt input, Linear(40, 12)	Training data: 2,417,199 gene-level pairs instead of 40-nt sites
Input encoding and base-pairing features (the authors' <code>get_cts</code> , <code>get_interaction_map_for_test</code>)	Training procedure: the training window of each pair is re-selected by the model before every epoch
Loss (CrossEntropy on the softmax output), Adam lr 1e-4, weight decay 1e-5, batch 256	Nine rounds: one epoch each in rounds 1-6, three in rounds 7-9
Test procedure: step 5, pair positive if any window > 0.5 (<code>decision_for_whole</code>)	

3. Code

`training/Mimosa.py` is not modified; the experiment only adds files, which call the authors' functions:

Path	Purpose
<code>training/mil_features.py</code>	Computes the authors' window features for every pair once (1.2 billion training windows) and stores them compactly (13 bytes per window)
<code>training/mil_select.py</code>	Scores all windows on GPU and keeps each pair's best window; also used for validation and testing (pair positive if best window > 0.5, i.e. <code>decision_for_whole</code>)
<code>training/mil_train.py</code>	Epochs on the selected windows with the original training loop
<code>training/mil_common.py</code> , <code>mil_metrics.py</code>	Feature storage helpers; gene-level metrics
<code>scripts/run_mil_*.slurm</code> , <code>submit_mil_scratch.sh</code> , <code>submit_mil_test.sh</code>	Slurm jobs; the rounds are chained with job dependencies
<code>experiments/composition/mil_shuffle.py</code>	miRNA swap test (Section 7)

Checks: the stored and rebuilt features are identical to the inputs `kmers_predict` builds (20/20 pairs, 7,996 windows), and the selection's gene-level decisions are identical to `kmers_predict` (20/20 pairs). The 120-epoch site model from the first report, run through the new path on the validation subset, gives metrics consistent with the original path on `test_unseen_pair` (accuracy 0.515 vs 0.514, specificity 0.148 vs 0.141).

4. Data

Use	Rows	Pairs	Windows
Training	<code>pairs_train_val.tsv</code> , split = train (2,380,213 gene-level + 36,986 40-nt sites; 50.6 % positive)	2,417,199	1,203.1 M (median 510 per pair)
Validation	every 9th row of split = val (the random 7 % validation split)	20,216	10.0 M
Test	<code>test_unseen_pair / source / target</code> , processable rows	42,476 / 379,727 / 394,110	21.5 M / 192.0 M / 199.4 M

Same prepared files as in the first report (`data/custom/`). Features: `data/custom/mil_features/` (11 GB on disk).

5. Training rounds

Round	Training window	Epochs	Train loss	Acc	PPV	Recall	Spec	F1	NPV
0	— (untrained model)	—	—	0.502	0.502	1.000	0.000	0.668	0.000
1	random	1	0.693	0.503	0.502	0.999	0.003	0.669	0.757
2	best (model of round 1)	1	0.675	0.508	0.505	0.990	0.023	0.669	0.693
3	best (round 2)	1	0.666	0.589	0.555	0.918	0.258	0.692	0.758
4	best (round 3)	1	0.588	0.733	0.732	0.736	0.729	0.734	0.733
5	best (round 4)	1	0.565	0.744	0.751	0.734	0.754	0.742	0.738
6	best (round 5)	1	0.561	0.746	0.747	0.746	0.746	0.747	0.745
7	best (round 6)	3	0.558 → 0.554	0.747	0.736	0.775	0.719	0.755	0.761
8	best (round 7)	3	0.549 → 0.547	0.756	0.753	0.764	0.747	0.759	0.759
9	best (round 8)	3	0.552 → 0.550	0.757	0.757	0.759	0.755	0.758	0.757

Validation metrics are gene-level (20,216 pairs). Selection took ~3.4 h per round on 6 GTX 1080 Ti + 2 RTX 4090 (rounds 2–6), ~8.5 h on the 1080 Ti alone (rounds 7–8) and ~6.4 h in round 9, when an RTX 4090 became free. One epoch takes 9 min on an RTX 4090 and 23 min on a 1080 Ti. Feature precomputation: ~1,000 CPU-hours, once. Loss values are not comparable across rounds, because the training windows change every round.

Round 1 learns nothing: a random window of a positive pair is almost never its binding site. From round 2 the model's own selections carry signal; rounds 3–4 show a sharp jump and rounds 4–6 plateau near 0.745. Rounds 7–9 gain about one point, and rounds 8 and 9 are equal within noise. **The round-9 model was chosen** (highest validation accuracy, recall ≈ specificity) **and tested**; the round-6 model had been tested earlier.

The training loss stays high because the original code applies softmax twice (inside the model and again in CrossEntropyLoss), which puts a floor of 0.313 on it even for a perfect model. It was kept for fidelity.

6. Gene-level test

Same test procedure and same pairs as in the first report.

Split	Model	Rows	n	Acc	PPV	AP	Recall	Spec	F1	NPV
test_unseen_pair	MIL, round 9	processable	42,476	0.767	0.759	0.705	0.789	0.745	0.774	0.776
	MIL, round 9	all rows	44,580	0.759	0.759	0.697	0.759	0.759	0.759	0.759
	MIL, round 6	processable	42,476	0.752	0.745	0.690	0.774	0.730	0.759	0.760
	Site model	processable	42,476	0.514	0.511	0.510	0.881	0.141	0.647	0.537
test_unseen_source	MIL, round 9	processable	379,727	0.771	0.762	0.710	0.796	0.746	0.779	0.782
	MIL, round 9	all rows	398,828	0.763	0.762	0.701	0.766	0.761	0.764	0.764
	MIL, round 6	processable	379,727	0.756	0.748	0.695	0.781	0.731	0.764	0.766
	Site model	processable	379,727	0.516	0.512	0.511	0.881	0.143	0.648	0.540
test_unseen_target	MIL, round 9	processable	394,110	0.763	0.759	0.702	0.777	0.749	0.768	0.768
	MIL, round 9	all rows	415,348	0.753	0.759	0.692	0.744	0.763	0.751	0.748
	MIL, round 6	processable	394,110	0.749	0.745	0.688	0.764	0.734	0.754	0.754
	Site model	processable	394,110	0.512	0.509	0.508	0.882	0.136	0.645	0.530

"Site model": the 120-epoch model of the first report. "all rows": unprocessable pairs counted as negative predictions. Results: runs/mil_scratch/test_round9/<split>/metrics.json (round 9) and runs/mil_scratch/test/<split>/metrics.json (round 6).

The MIL model is balanced (recall ≈ specificity), and its results are the same for the three kinds of generalisation (new miRNA, new target, both new) and match the validation subset. Round 9 is 1.4–1.5 points more accurate than round 6 on every split, as on validation (+1.1). Its accuracy being close to the paper's 0.757 on miRAW is a coincidence: the benchmarks differ.

7. Does the decision depend on the pair? (miRNA swap)

Same test as in the first report: targets are kept and miRNAs are swapped between positive and negative pairs (150 targets per group). On our data the test uses test_unseen_pair, whose miRNAs and targets were never seen in training.

Target	miRNA	MIL (round 9), our data	MIL (round 9), miRAW	Authors' model, miRAW (first report)
From a negative pair	its own	0.293	0.180	0.413
From a negative pair	taken from a positive pair	0.220	0.280	0.947
From a positive pair	its own	0.840	0.453	0.933
From a positive pair	taken from a negative pair	0.193	0.180	0.387

Share of pairs called positive. Swapped pairs have no known label; random miRNA–target combinations are expected to be mostly non-interacting.

On our data a positive target keeps its high call only with its own miRNA: with another miRNA it falls to the rate of negatives (0.840 → 0.193), and a miRNA taken from a positive pair does not raise a negative target (0.293 → 0.220). The decision depends on the miRNA–target pair, not on either one alone, which is the opposite of the authors' model on miRAW. On miRAW the MIL model behaves the same way apart from a small miRNA effect (0.180 → 0.280); it is more conservative there (positives called at 0.453), as expected from a model trained on a different benchmark. The round-6 model gave the same picture on our data (0.807 → 0.233 and 0.267 → 0.227).

8. Limitations

- The training procedure differs from the paper (gene-level data, window re-selection between epochs): these are results for Mimosa's architecture with multiple-instance training, not a reproduction.
- Single run; the variance across random seeds is unknown.
- Round 1 uses a random window. Rounds 7-9 (three epochs each) were added because the loss was still falling after round 6. Two MIL models (rounds 6 and 9) were tested; round 9 was chosen on validation before testing.
- The swap test uses a sample of 150 targets per group, not a full evaluation.

9. Conclusions

- With the same architecture, features and test rule, multiple-instance training on all gene-level pairs gives a balanced model: accuracy 0.76-0.77, recall 0.78-0.80 and specificity ≈ 0.75 on all three test splits, against accuracy 0.51-0.52 and specificity 0.14 with the paper's training method.
- Its decisions depend on the miRNA-target pair, not on miRNA identity, also for miRNAs and targets never seen in training.
- Since the architecture is unchanged, the gap between the two reports comes from the training data and procedure.